Anirudh Kamath

anirudhkamath.github.io

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Education

- **University of Utah** — Salt Lake City, UT
  Master of Science in Computer Science — *Aug. 2023 – present*
  Courses: Graduate Algorithms, Programming Languages, Advanced Computer Networks
- **Manipal Institute of Technology** — Manipal, India
  Bachelor of Technology in Computer and Communication Engineering — *Aug. 2016 – Aug. 2020*
  Courses: Machine Learning, iOS/Android Mobile App Development, Network Programming, IoT

Skills

- **Software engineering**: Python, bash, Django, NGINX, Redis, PostgreSQL, GraphQL, Ansible
- **DevOps**: Jenkins, Docker, docker-compose, Kubernetes, pytest
- **Cloud**: AWS – EC2, ECS, ECR, Lambda, S3, CloudFormation, RDS, SSM, Systems Manager
- **Telemetry**: Telegraf, Prometheus, Thanos, Alertmanager, Grafana
- **Developer tools**: Git, Bitbucket, GitHub, JIRA, make, invoke

Work Experience

- **Baxter** — Bengaluru, India

  Network Automation Engineer — *Sep 2020 - Jul 2023*

  - Built and maintained containerized **Python** network automation applications and **REST API** services with frameworks such as **Django** hosted on **AWS Elastic Container Service** to manage network data and automate network operations for **4000+ network devices**

    Skills: ECS, EC2, CloudFormation, S3, RDS, Python, REST APIs, GraphQL, Django, PostgreSQL
  - Developed a **distributed telemetry** system that collects and stores **1M network metrics / 1.4B data points** every day from the Baxter network in a **time-series database** consumed through **Grafana** dashboards and alert notifications produced by **Alertmanager** – bringing down MTTR to a matter of minutes

    Skills: SNMP, DNS, HTTP, TLS, NGINX, Prometheus, Thanos, Telegraf, Grafana, Alertmanager
  - Developed internal tooling to **improve developer experience** within the network automation team, examples of such tools I built:
    - **Infrastructure as Code (IaC)** tool to deploy **Ansible AWX**[1] resources via a **Jenkins** pipeline
    - **Telegraf[2] configuration generator** and testing playbook using **Ansible** and **Jinja2** templates
    - Sandbox support for all network automation applications using **docker-compose** and **invoke, make, and bash scripts** to set up containerized sandboxed environments for developers building network automation features

    And provided developer documentation for the same maintained in Markdown

    Skills: IaC, Jenkins, Ansible AWX, Jinja2, Docker, docker-compose, invoke, make, bash

  Intern — *Jan 2020 - Aug 2020*

  - Delivered an automation workflow using **Ansible** to provision **200 Cisco UCS** E-Series servers with **ThousandEyes** endpoint agents and in-house developed monitoring agents to monitor network level metrics for internal web resources globally – HTTP, DNS, SMTP, and other core services

    Skills: Linux, Docker, Ansible, Jinja2, HTTP, DNS, SMTP
  - Developed custom NAPALM[3] drivers for gathering data from Baxter network devices and **Python plugins** that used this data to populate our network source of truth **Django application** – delivered a unit testing framework for the custom NAPALM drivers developed using **pytest**

    Skills: Python, Python-poetry, pytest, Markdown for docs

Projects

- **ReachUs**: Web app to bridge the gap between Baxter employees and company leadership during COVID-19. Powered by Django, CSS, Javascript
- **Network automation blog[4]**: A collection of articles on technologies in the network automation space

Certifications

- Cisco Certified Network Associate (CCNA)
- Cisco Certified DevNet Associate (DEVASC)
- AWS Certified Cloud Practitioner (CCP)

References

[1] **AWX Project by RedHat**: https://www.ansible.com/products/awx-project/

[2] **Telegraf by InfluxData**: https://github.com/influxdata/telegraf

[3] **NAPALM**: https://github.com/napalm-automation/napalm

[4] **network-automation-blog**: https://anirudhkamath.github.io/network-automation-blog/